HOW TO INSTALL A FLYBACK TRANSFORMER ON A G07 MONITOR

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It will be assumed that you are aware of the dangers of removing a chassis and the need to discharge the tube at the suction cup (anode) and that you are familiar with soldering and working with hand tools. The author will not assume any liability for use of or an inability to competently use these instructions. This information is to be used as a guide only. Research other threads, posts, and YouTube videos to absorb as much information as needed to successfully complete this task.

You will need the following:

- High-quality soldering iron
- Desoldering wick and/or a solder sucker
- Assorted hand tools (screwdrivers, diagonal cutters, wire strippers, etc.)
- A replacement flyback transformer
- Camera (if you want to document your work/experience)

Important! Turn off power to the monitor and discharge it before performing any of the following steps!

First, you will need to remove the existing flyback. Start by removing all the solder from the existing mounting holes. This first image shows where a plunger type solder sucker was used to remove a large portion of the existing solder and solder braid is used to clean between the pad and the legs of the flyback.

Figure 1 - Flyback Mounting Holes

Be careful if you are wiggling the flyback side to side to help remove solder or remove the flyback. You will not want to crack the PC board or pull a pad off the board.

If you have desoldered the flyback and you cannot pull it all of the way off the board then look for two screws on the side where the flyback may be mounted to a metal plate.

Now you should be able to remove the old flyback. If you note in this picture we are removing a recent installed flyback. (We were surprised to see this flyback fail so early in its lifespan.)

You may now notice that there is a wire from the flyback that is attached to the one of the two control knobs. We will need to pull back the protective sleeve to expose the bare wire.

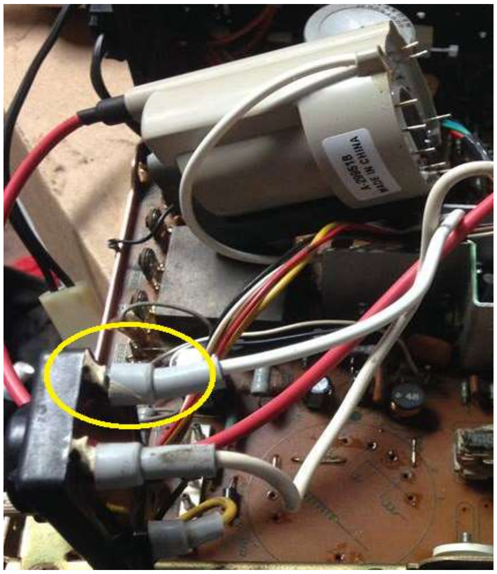

Figure 2 – Protective Sleeve

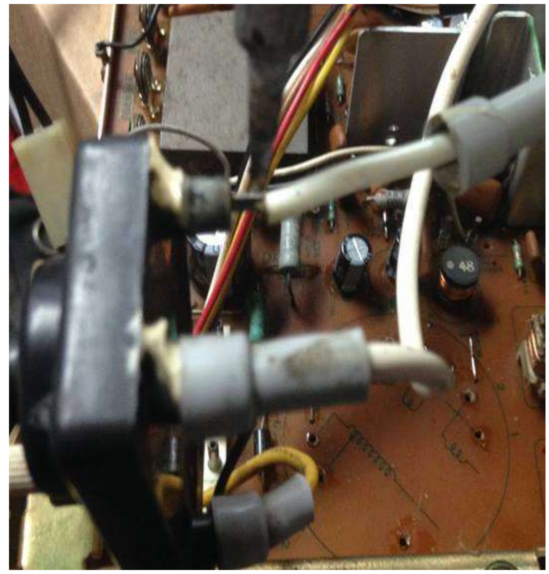

Figure 3 – Protective Sleeve Removed

Use the soldering iron to heat up the connection and pull on the wire to remove it. If you have issue then cut the wire and then continue to remove the connection. Once the wire is removed, use a solder sucker or desoldering braid to remove remaining solder from the hole.

Before you go any further take the protective sleeve and place it on the wire of your new flyback and use some tape to hold it there until you are ready to make the connection to the control. You can set this off to the side but it is common for people to solder the wire on and then realize they forgot the protective sleeve and then have to remove it and do it all over again.

To complete removal of the flyback, grasp the back of the anode cap where the and squeeze firmly. The anode cap fastens to the monitor via a clip whose prongs are inserted into the anode hole on the monitor. Squeezing the anode cup in this fashion should bring the wire clip's prongs closer together making it easier to remove the anode cup from the monitor. Move the cap from side to side to dislodge the prongs and remove the cap. Try to avoid scratching the glass if you can.

This next photo shows that the flyback's thru-holes are clear of any solder - you can see right through them.

You will want to make sure that the holes are as clean as possible before installing the new flyback transformer.

Alcohol and Q-Tips were used to remove the gummy residue from around the solder pads of the flyback. If you have flux remover, use it. But isopropyl alcohol works in a pinch.)

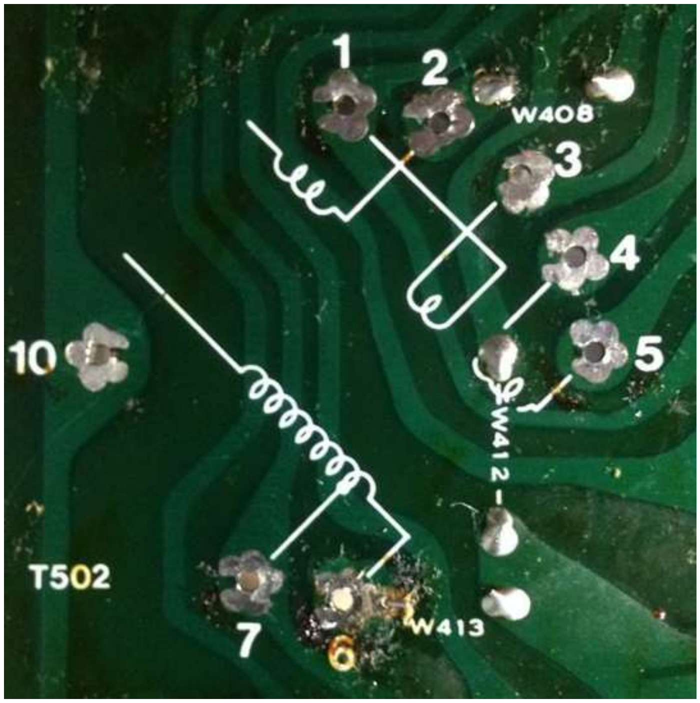

Figure 4 - Thru-Holes Cleaned

You may sometimes have problems installing the new flyback as the legs may be too thick to fit through the existing holes. Using a utility knife you can scrape the legs clean, and also use it to remove any solder residue on the legs.

It appears that you may find that some flybacks have been pre-tinned to make it easier to solder them to the board and sometimes the solder on the legs may be too thick.

Figure 5 - Scraping/Cleaning Flyback Legs

Line up the legs of the flyback to the holes on the mounting side of the PC board. You may find installation of the flyback to be easier if you remove metal bracket that the HOT (**H**orizontal **O**utput **T**ransistor) is mounted to. This will make it easier to see the holes as you try to mount the new flyback.

To remove this bracket, remove the two screws show in the first photo below. there is one additional screw shown in the second photo.

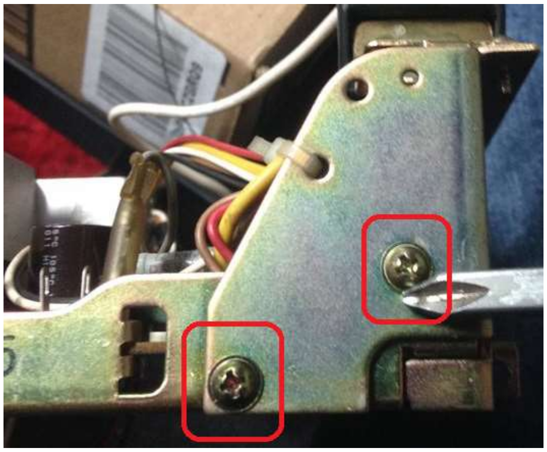

Figure 6 – Protective Sleeve

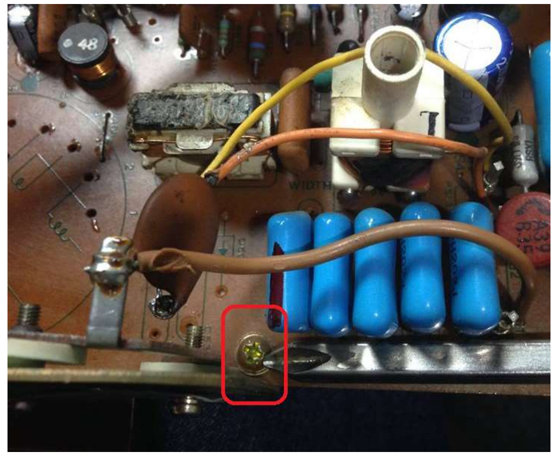

Figure 7 – Protective Sleeve Removed

If the bracket does not come off right away there is another screw below the two controls. You do not need to remove the screw all the way but perhaps by giving it a few turns and allow for some play the bracket will come off.

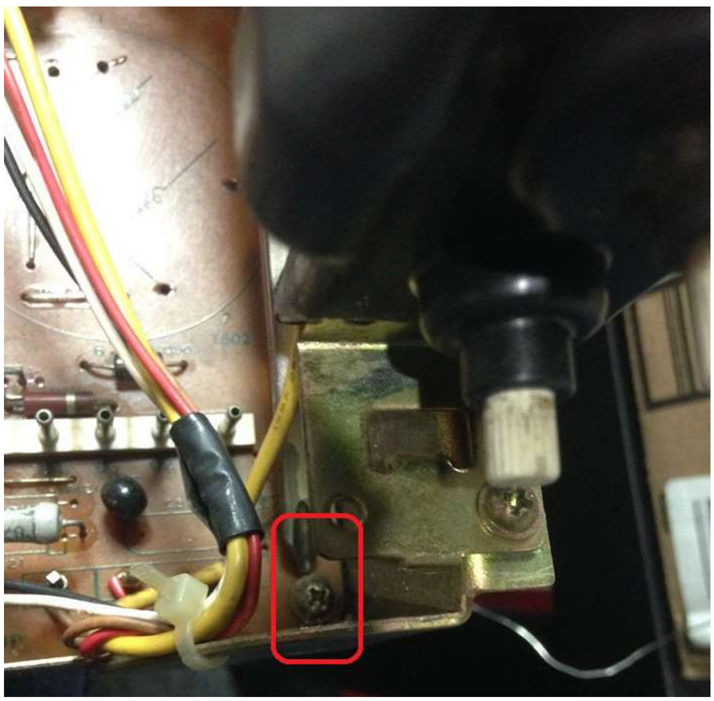

Figure 8 - Just a Few Turns...

The next photo shows the bracket is now out of the way but the wires attached may make it cumbersome to handle while holding the PC board and the flyback. But now you can see the holes where the flyback is to be mounted.

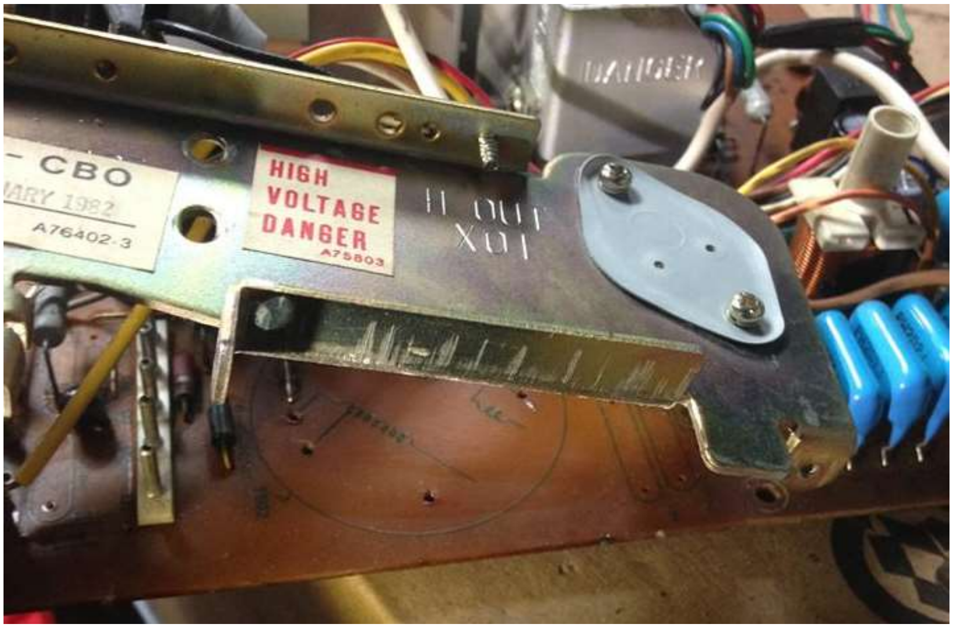

Figure 9 - Clear View of the Mounting Holes

Now (finally!) install the flyback. You may find that the flyback might not mount flush or at all. The legs may be pointing in different directions and may not line up with the holes. You may have to put some legs in one part of the flyback in to the holes and get them lined up and then move to a different set of leg and get them to line up with the holes and then slowly work your way around until you figure what legs need to be bent (and where) to line up with the holes.

When you go to mount the flyback start with the legs that are the farthest away from the HOT. This will allow you to look under the flyback as you try to install it and will allow you take a small screw driver to pry the last couple of legs in order to get them to line up with the holes.

Figure 10 - Aligning the Flyback Pins

You will want to make sure the flyback is fully flush with the board before you start to solder it in. Start by soldering two legs at opposite sides to help hold it in place while you solder the remaining pins. I would recommend crossing back and forth from one side to the other as you solder the legs. The idea here is to avoid concentrating heat all in one area.

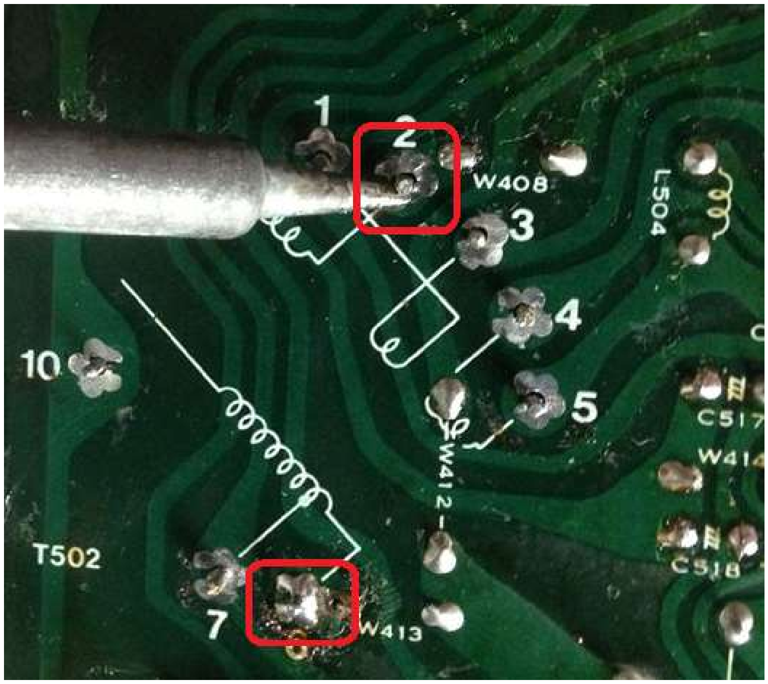

Figure 11 - Soldering Opposite Ends

Let one area cool while you go to the other side and solder and work your way back and forth around the legs of the flyback. When you are done soldering the leg then use alcohol and Q-Tips (or flux remover) to clean the pads to remove any remaining flux residue.

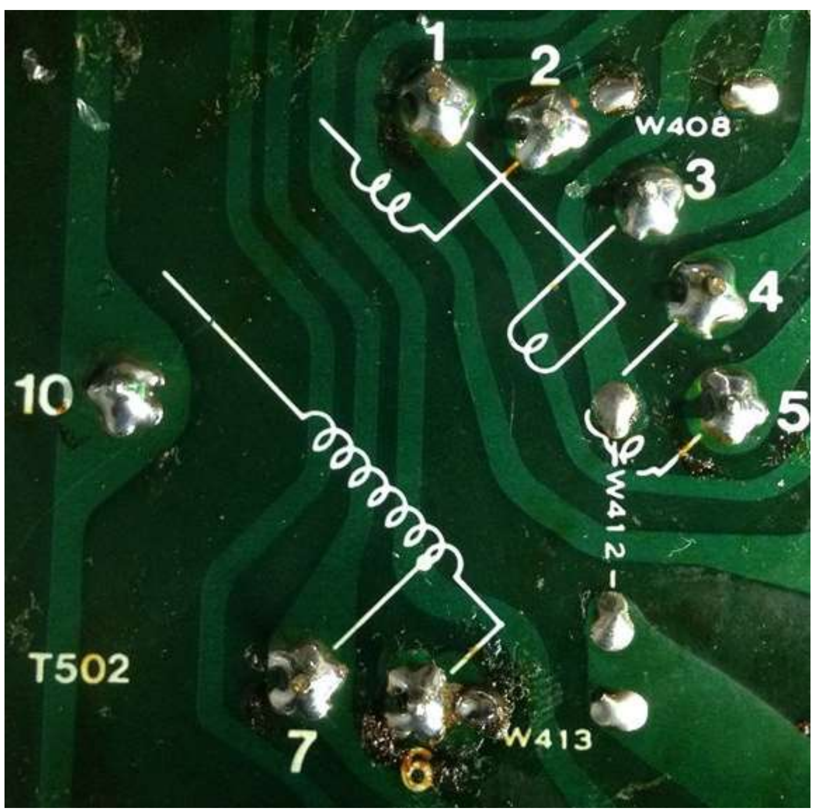

Figure 12 - New Work, All Cleaned

Put the metal bracket back on to the board and reinstall the two screws on one side and the one screw below the HOT and tighten that one screw below the controls.

Now solder the wire from the flyback to the control. Do not forget to put the protective sleeve on to the wire before you solder.

If the hole is not exposed on the control connection then use braid or a solder sucker to remove the excess solder. When the hole is open bend the tip of the wire in to an L shape. (Before connecting the wire, make sure the protective cover is on the wire and is facing the correct direction!)

Figure 13 - Reinstalling Wire

Insert the wire into the hole and using needle nose pliers pinch the tip of the wire to make a good mechanical connection and then solder. I pull the wire straight as I remove the solder iron and allow it to solder joint to cool before I let go of the wire. Slide the protective cap over the connection.

When flybacks fail they will often take out fuse F901. You will want to find this fuse. It is located by the big capacitor (huge round can). The fuse could be the type that is soldered in to the board ("pigtailed") or there could be a fuse holder allowing you to remove the blown fuse and install a new fuse. It is possible that a fuse holder was installed on the chassis in another location with wires going back to where the fuse is located on the board. This fuse is a 1.25-amp fuse.

Figure 14 - Location of F901 (Pigtail Fuse)

Reinstall the chassis and be prepared to measure the B+ before putting the monitor back in to operation.